



GREEN ALIBI

Complete Figure Set — all graphs, maps, and plots referenced in the Research Paper and interactive dashboard, compiled in one document.

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Testing the SIF-vs-NDVI drought-detection lag, Marathwada, Maharashtra
2015 & 2018 (drought years) · 2020 (normal monsoon year)

1. Study Area

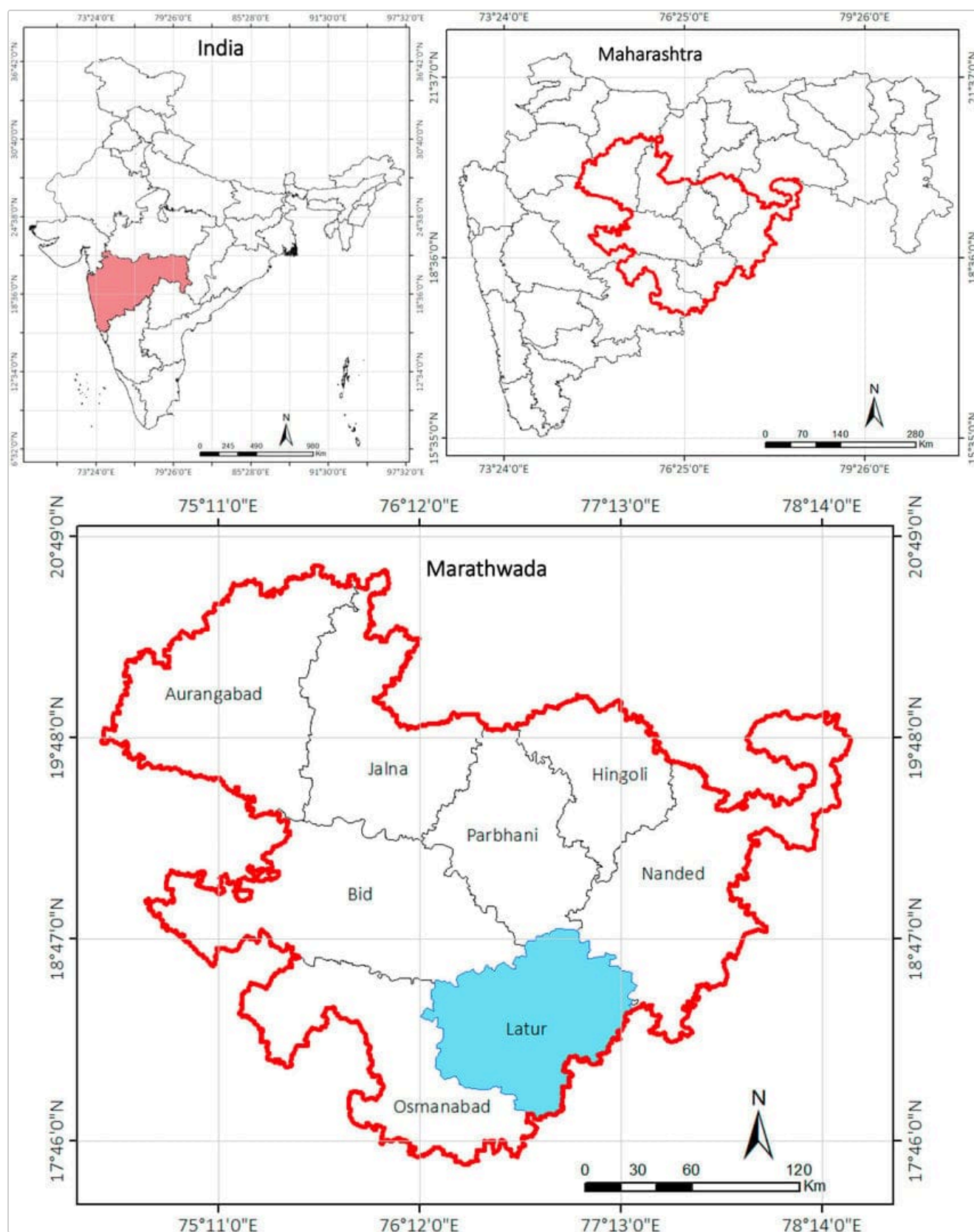


Figure 1

Location of the study area — (a) Maharashtra within India, (b) Marathwada within Maharashtra, (c) the eight constituent districts of Marathwada.

2. Physical Basis — Fluorescence & NDVI

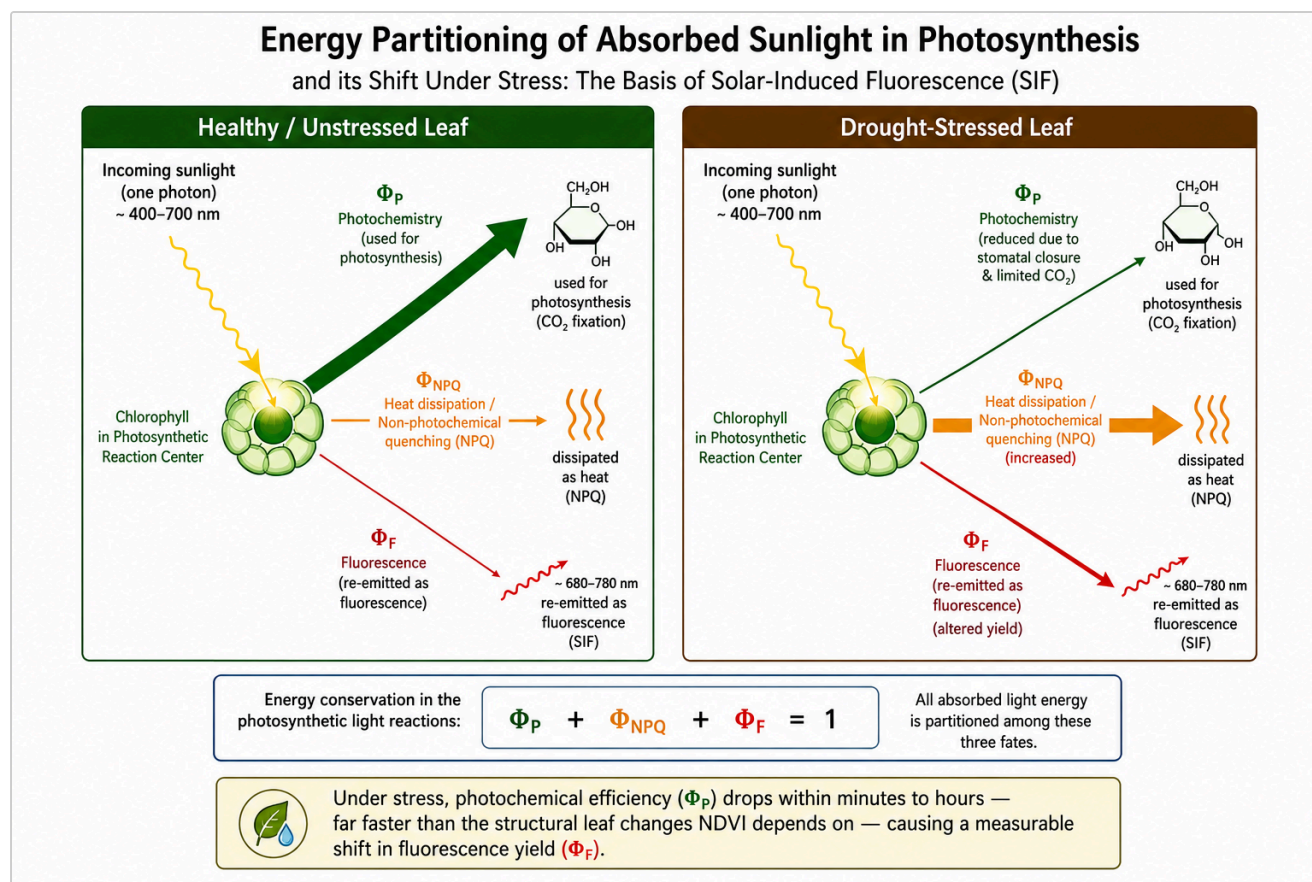


Figure 2

Energy partitioning of absorbed sunlight in photosynthesis, and its shift under drought stress — the physical basis of Solar-Induced Fluorescence.

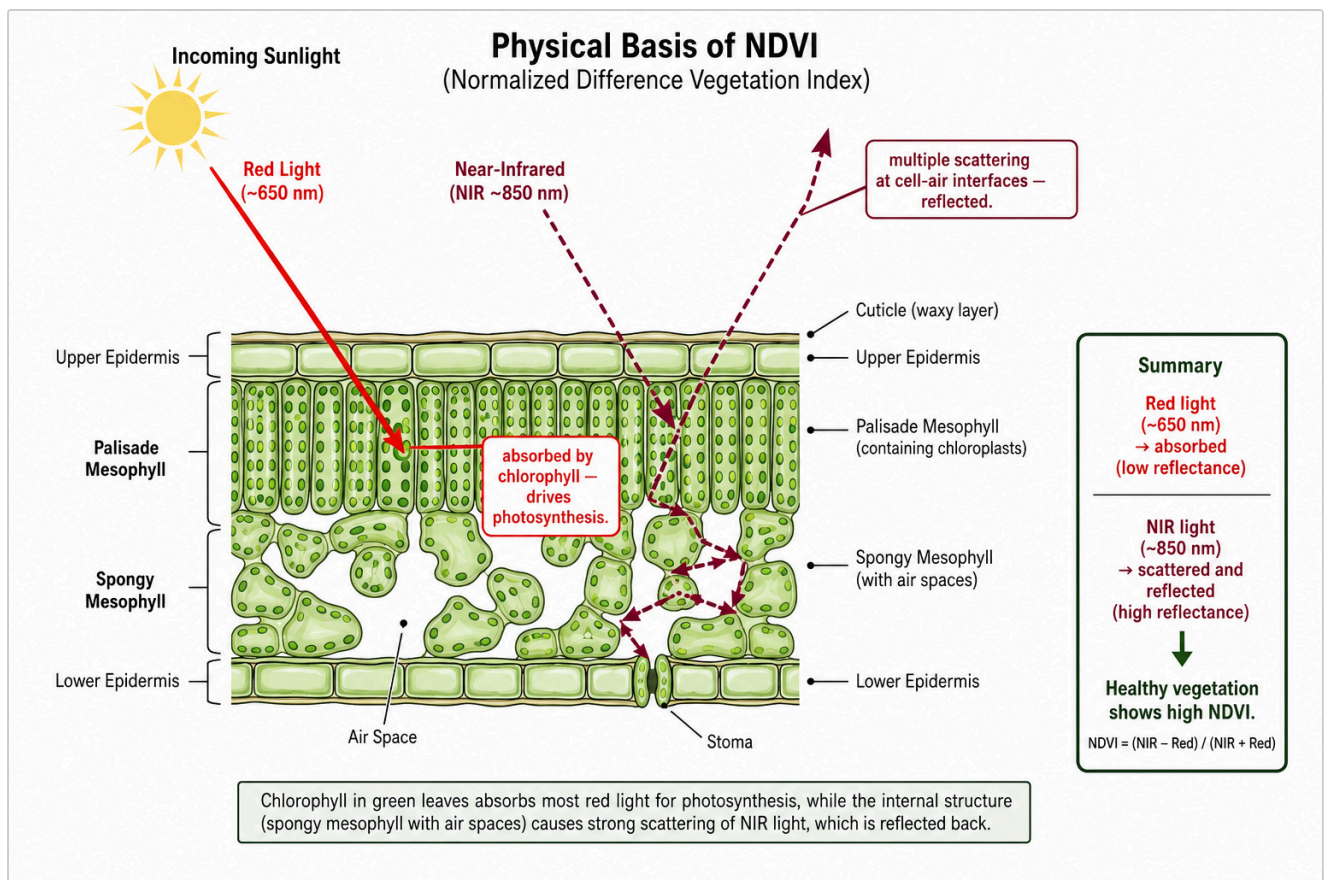


Figure 3

The physical basis of NDVI: red light absorption by chlorophyll in the palisade mesophyll versus multiple scattering and reflection of near-infrared light in the spongy mesophyll.

3. Data & Methodology — Boundary Verification

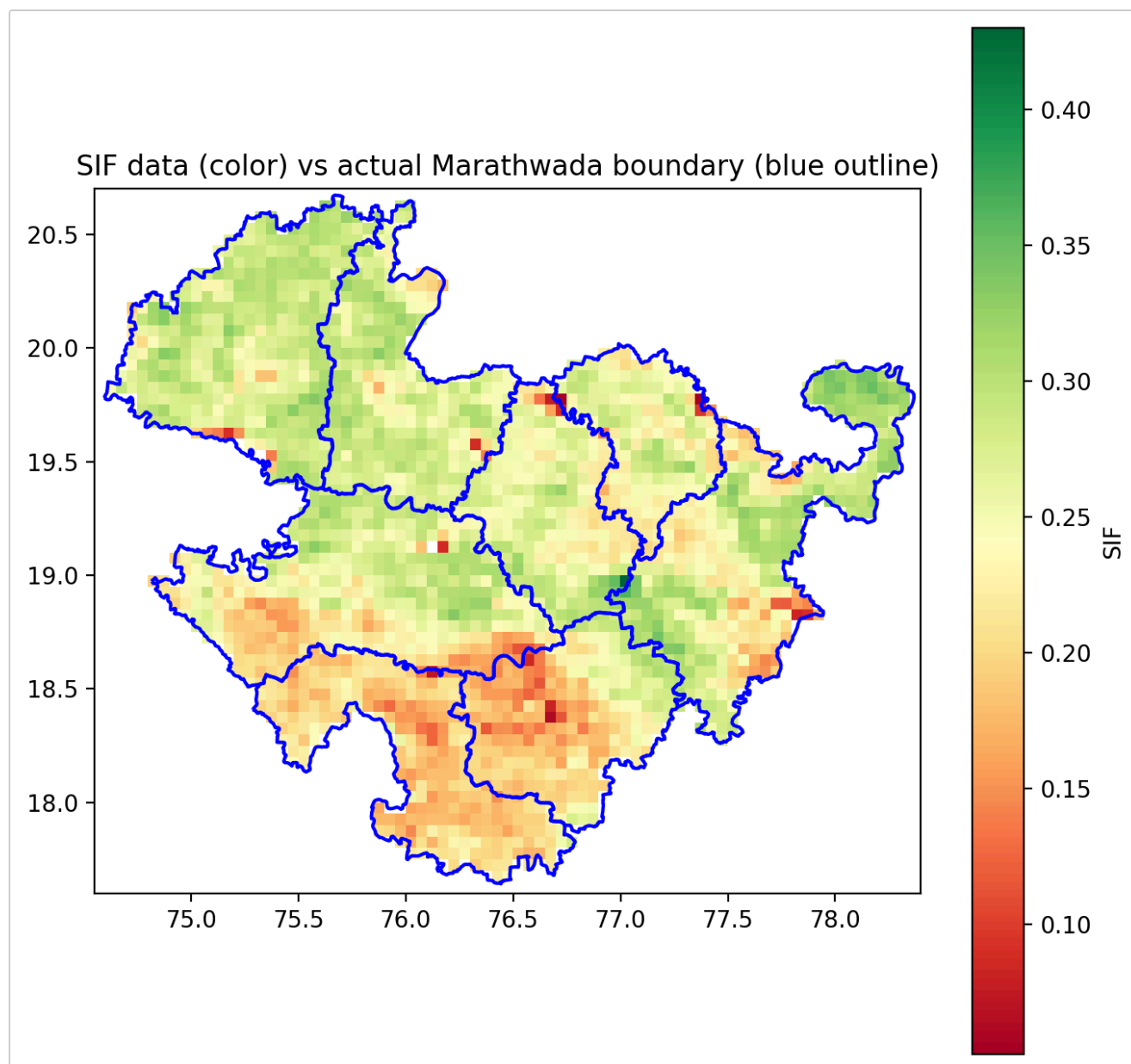


Figure 4

Verification: clipped SIF data (color) overlaid with the true Marathwada boundary (blue outline) — data is confined exactly within the real district shapes.

4. Seasonal SIF vs. NDVI Trajectories

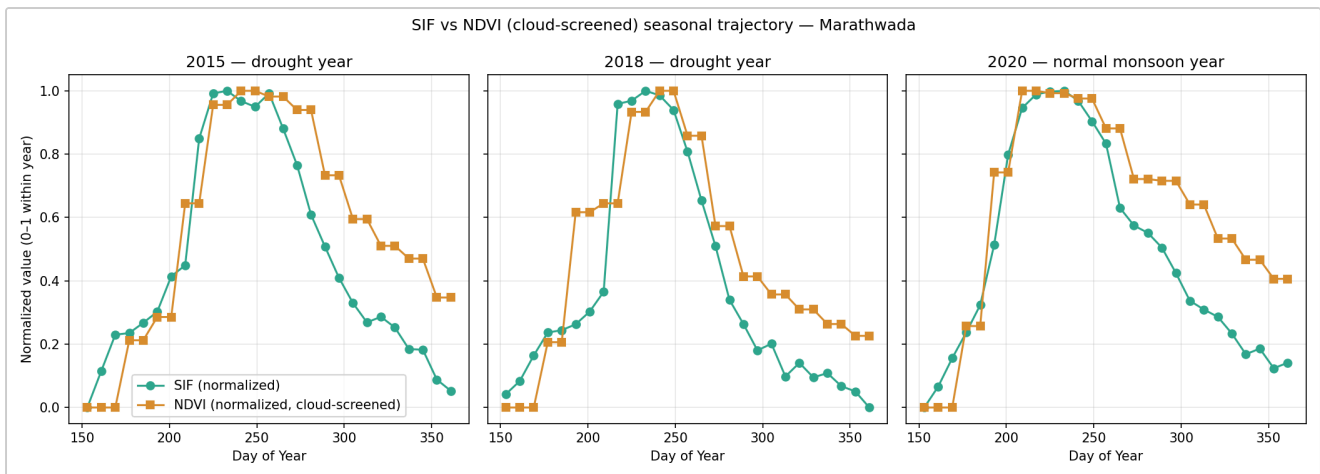


Figure 5

Normalized SIF and NDVI seasonal trajectories, Marathwada, 2015 (drought), 2018 (drought), and 2020 (normal monsoon). SIF's post-peak decline precedes NDVI's decline in all three years.

5. Quantitative Lag Analysis

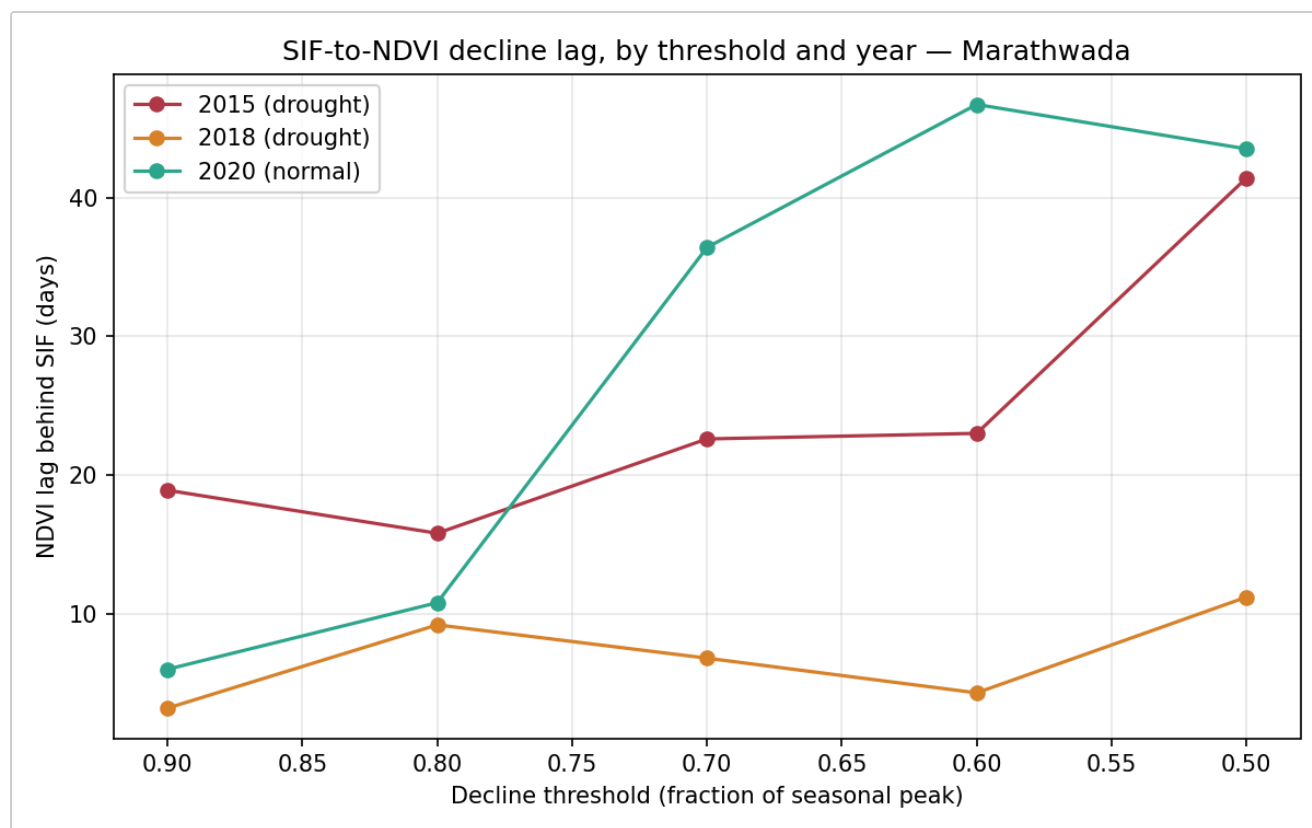


Figure 6

SIF-to-NDVI decline lag (days), by decline threshold and year.

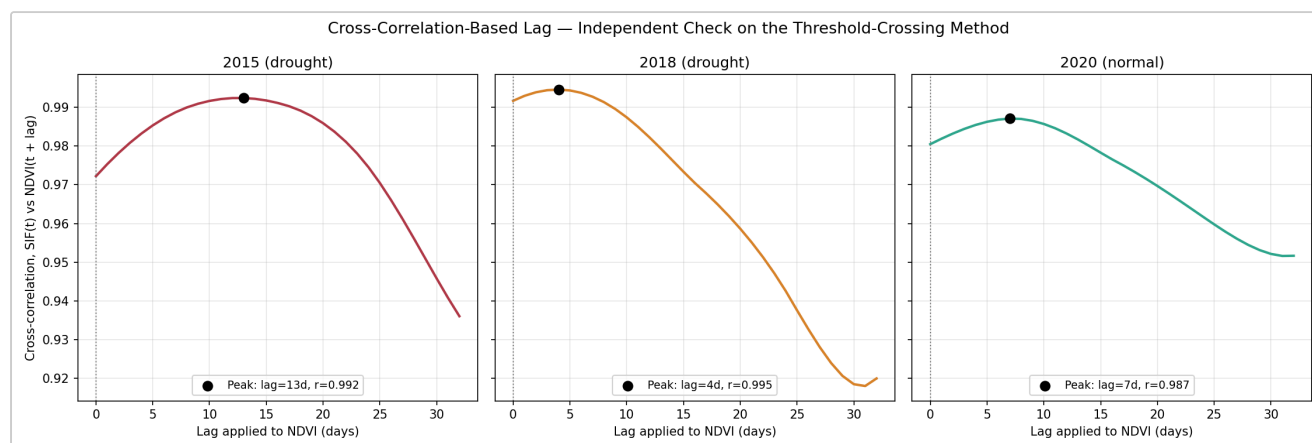


Figure 7 — Robustness check (independent method)

Cross-correlation between SIF(t) and NDVI(t + lag), by year. Lags tested are capped at N/4 of the decline window (a standard guideline) to avoid boundary-pinned, unreliable peak estimates. All three peaks shown are interior maxima, well clear of that boundary.

Honest reporting note: the cross-correlation method independently confirms that SIF leads NDVI in all three years (strengthening H1), but it does not reproduce the same year-to-year lag ranking as the threshold-crossing method above — reported as a method-sensitivity finding in Research_Paper.md §4.6, rather than smoothed over.

6. Spatial SIF Analysis

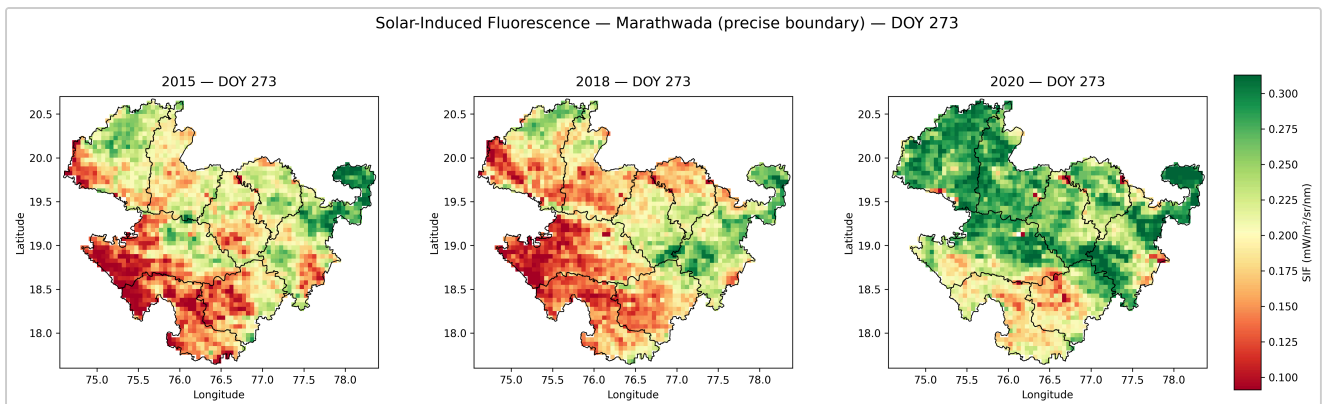


Figure 8

Spatial SIF distribution, day-of-year 273, 2015/2018/2020, masked to the precise Marathwada boundary. Boundary outline drawn in black.

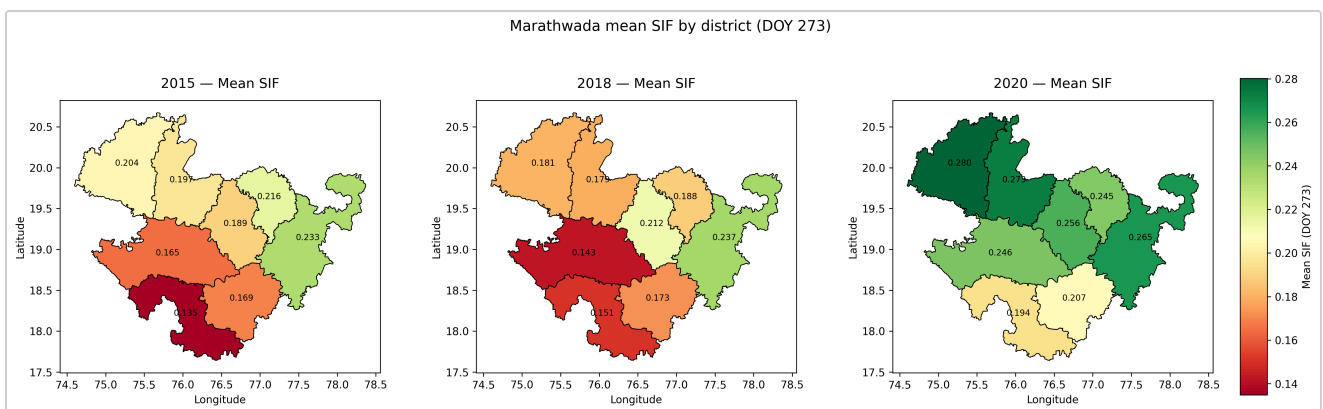


Figure 9

Mean SIF by district, day-of-year 273, 2015/2018/2020.

6a. Spatial SIF — Interactive District Map (static export)

The dashboard's Folium interactive map, exported here as a static image per year for this compiled document. Deeper red = lower mean SIF (more stress).

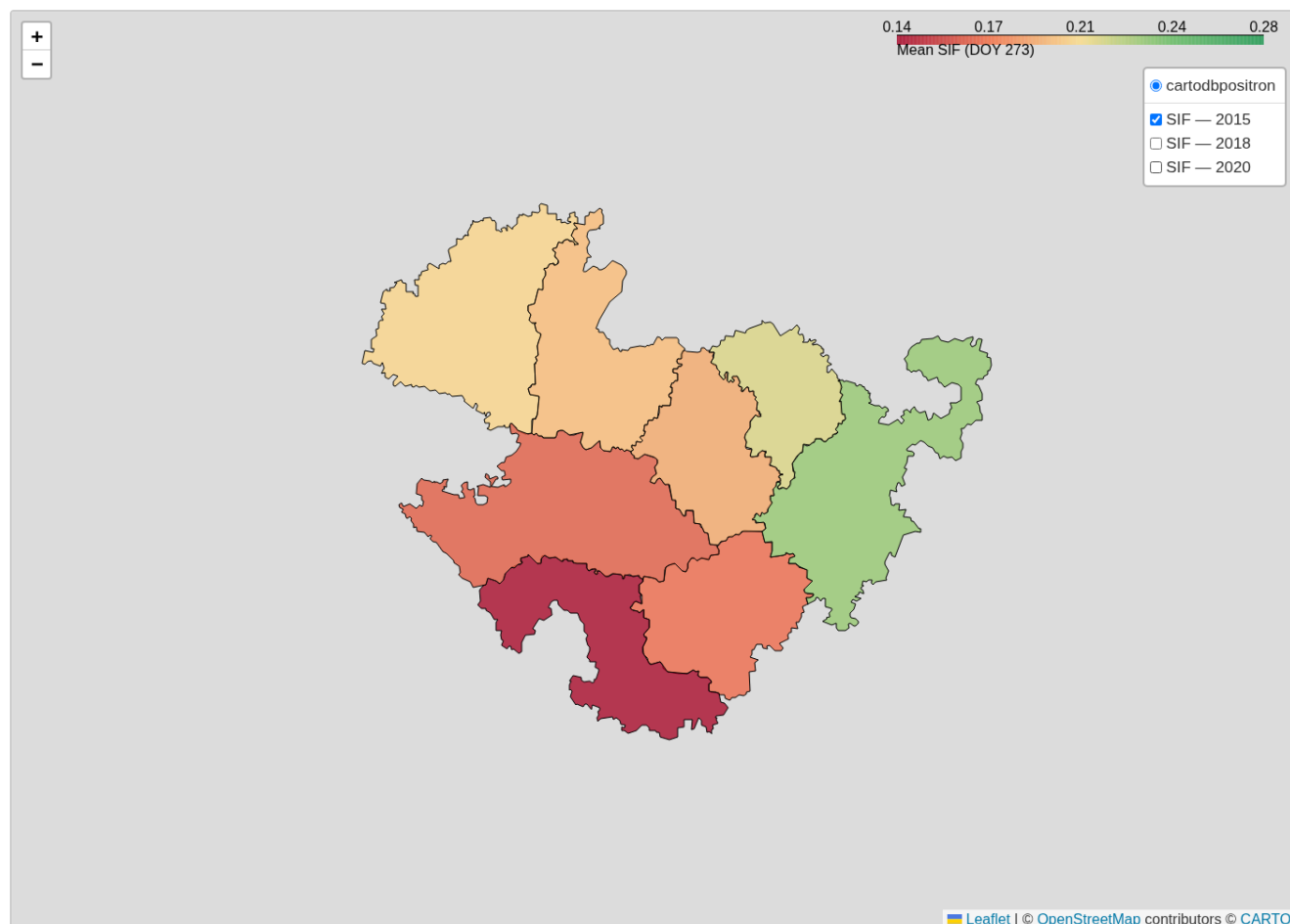


Figure 10a — SIF by district, 2015 (drought)

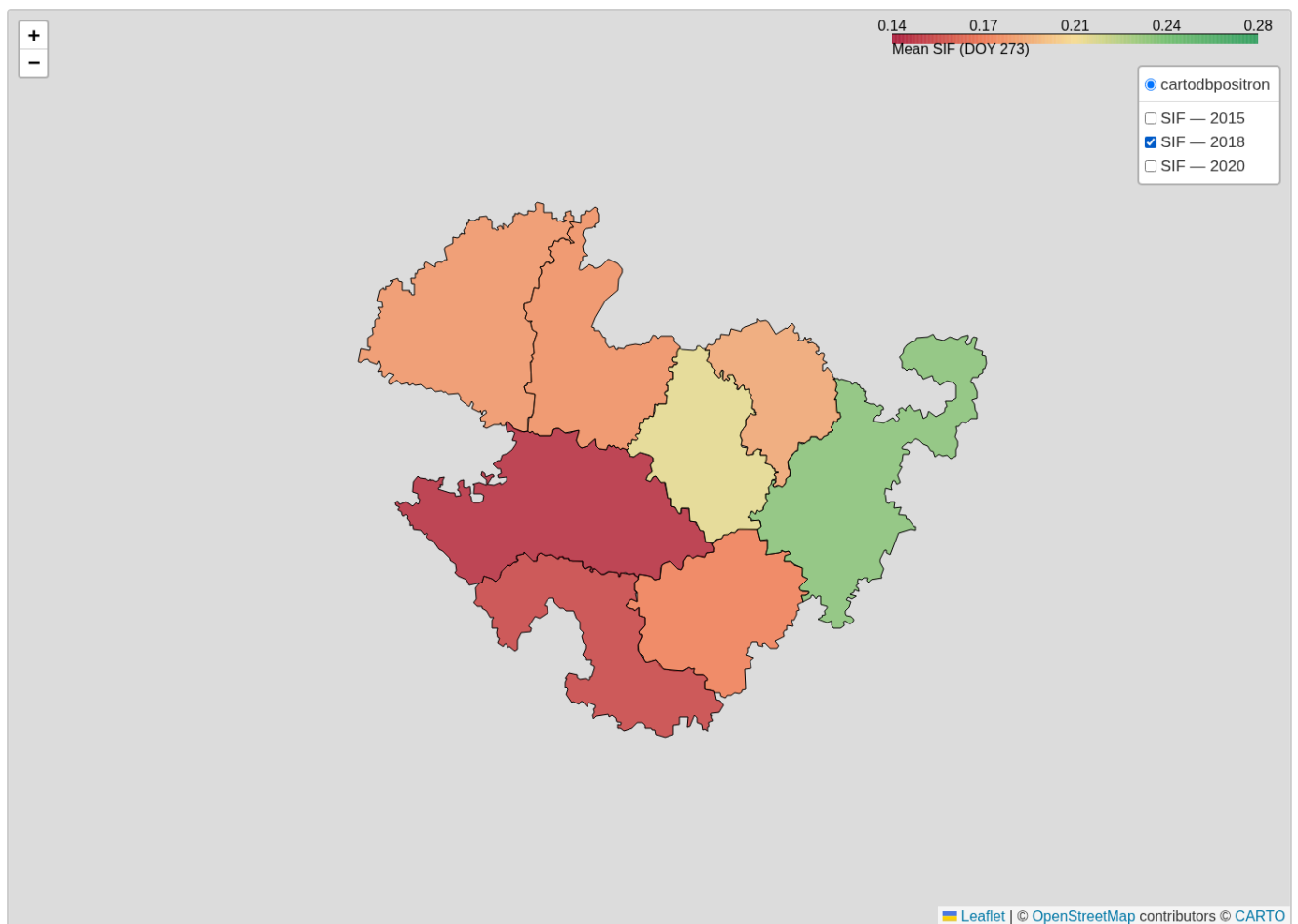


Figure 10b — SIF by district, 2018 (drought)

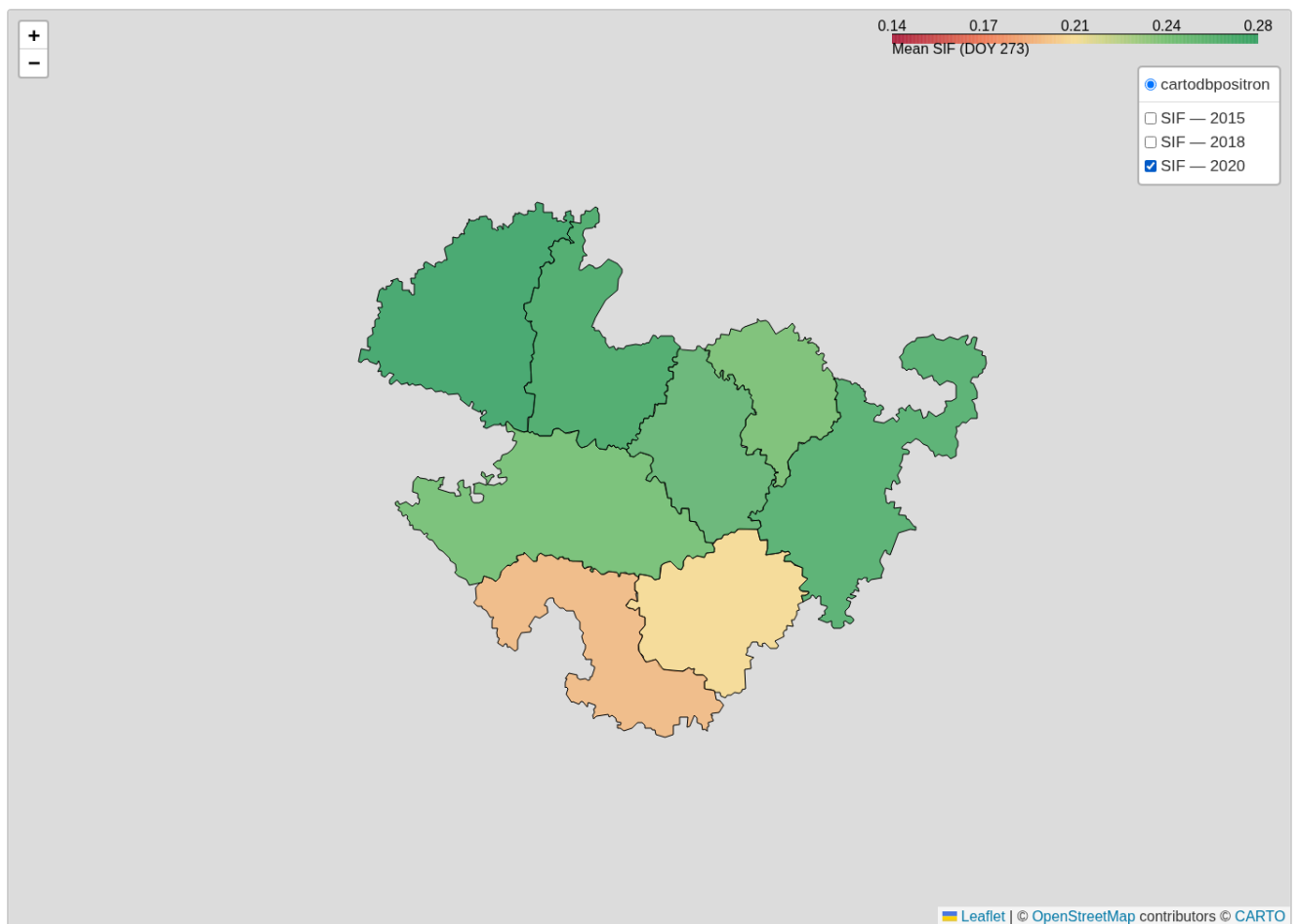


Figure 10c — SIF by district, 2020 (normal)

7. Rainfall Validation

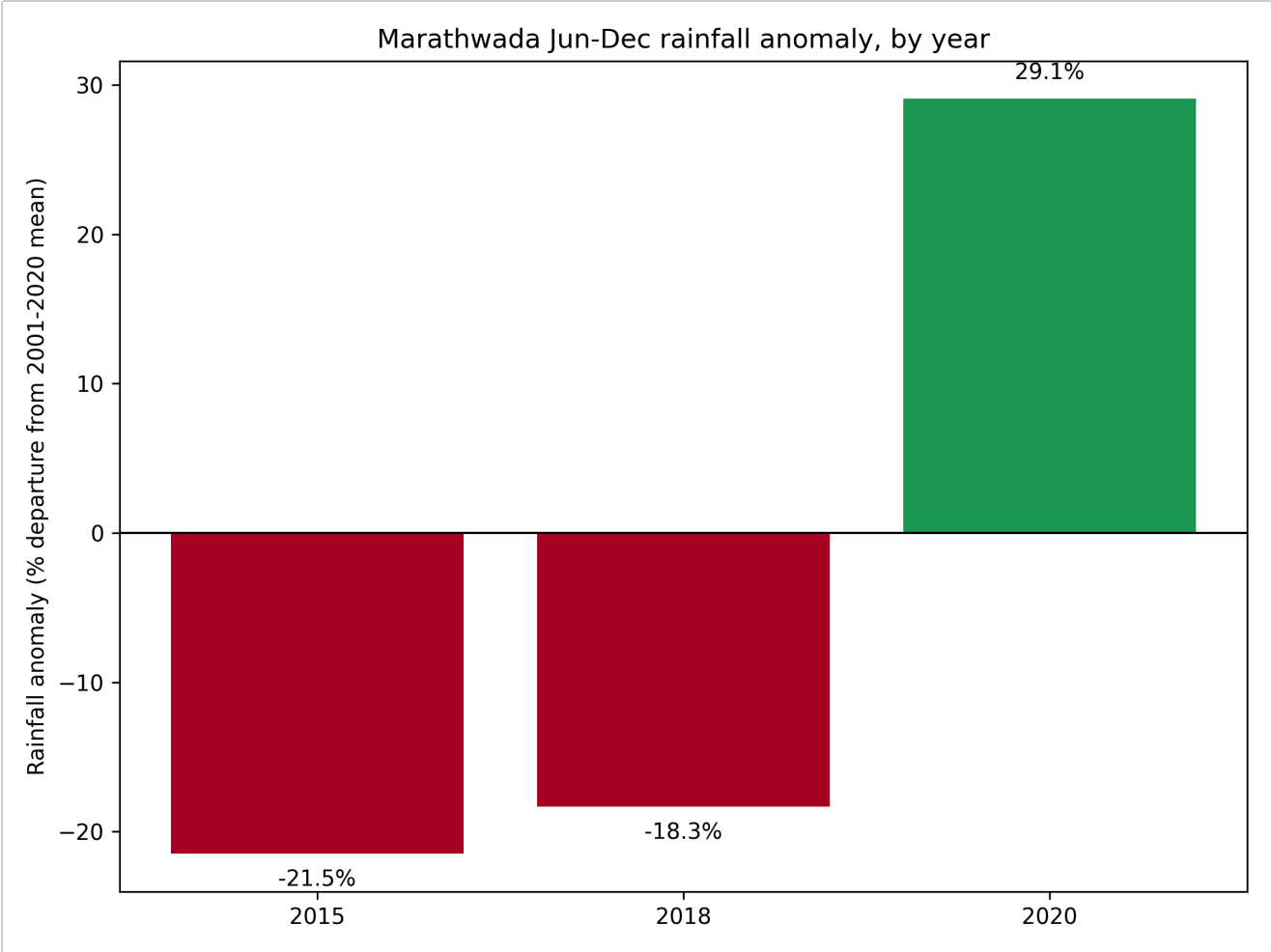


Figure 11
Regional rainfall anomaly (% departure from 2001–2020 climatological mean), by year.

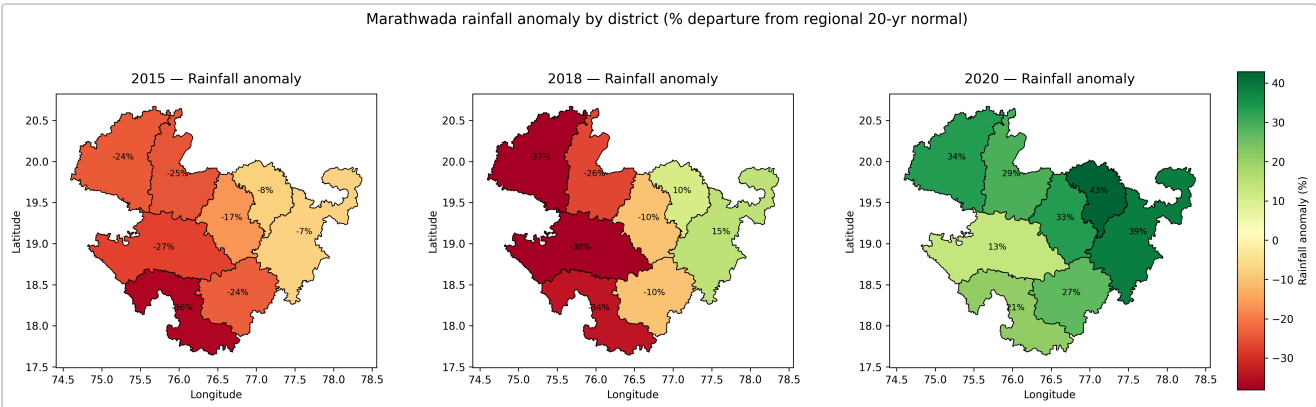


Figure 12
Rainfall anomaly by district (% departure from regional 20-year normal), 2015/2018/2020.

7a. Rainfall Anomaly — Interactive District Map (static export)

The dashboard's Folium interactive rainfall map, exported here as a static image per year. Deeper red = larger rainfall deficit.

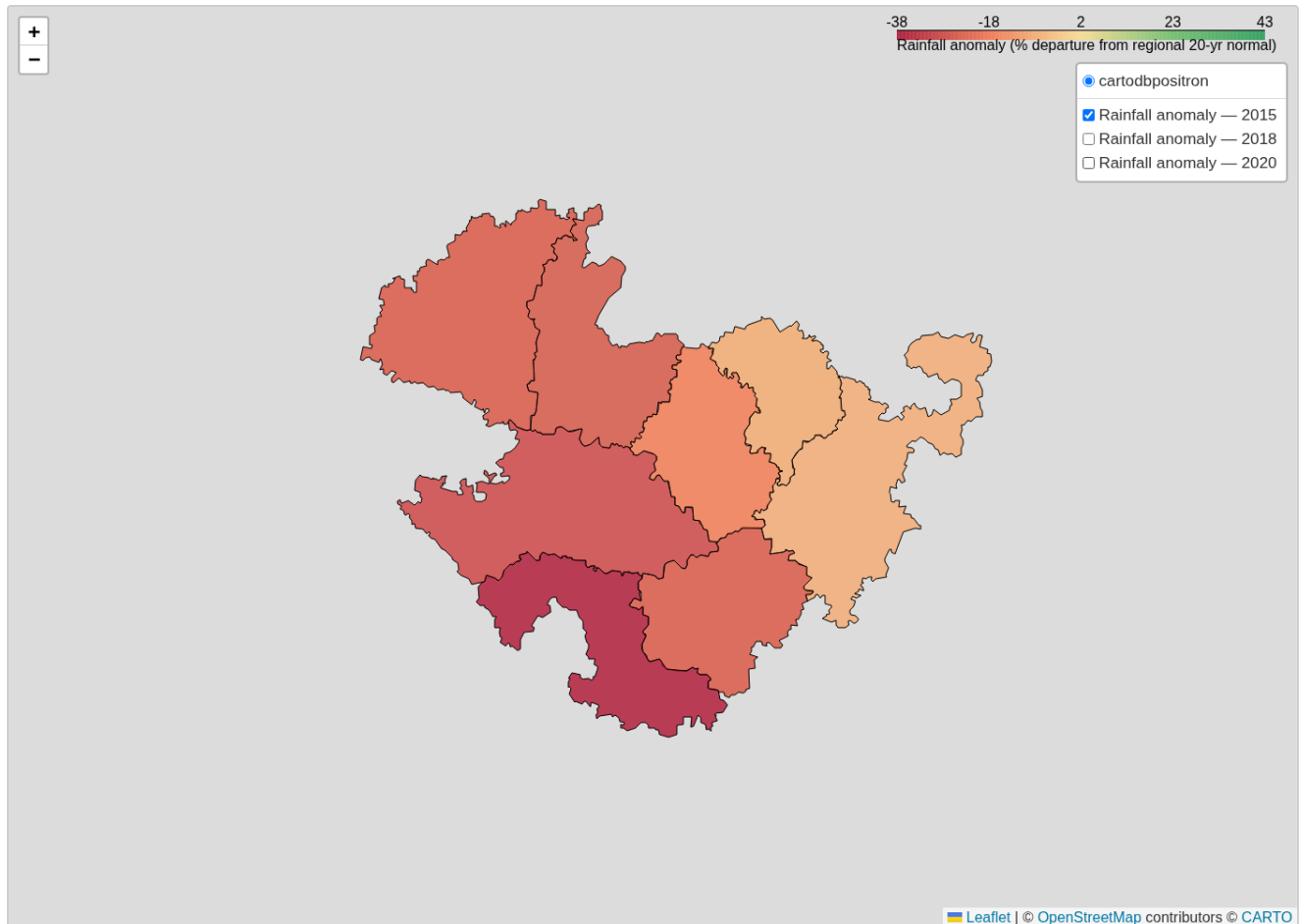


Figure 13a — Rainfall anomaly by district, 2015 (drought)

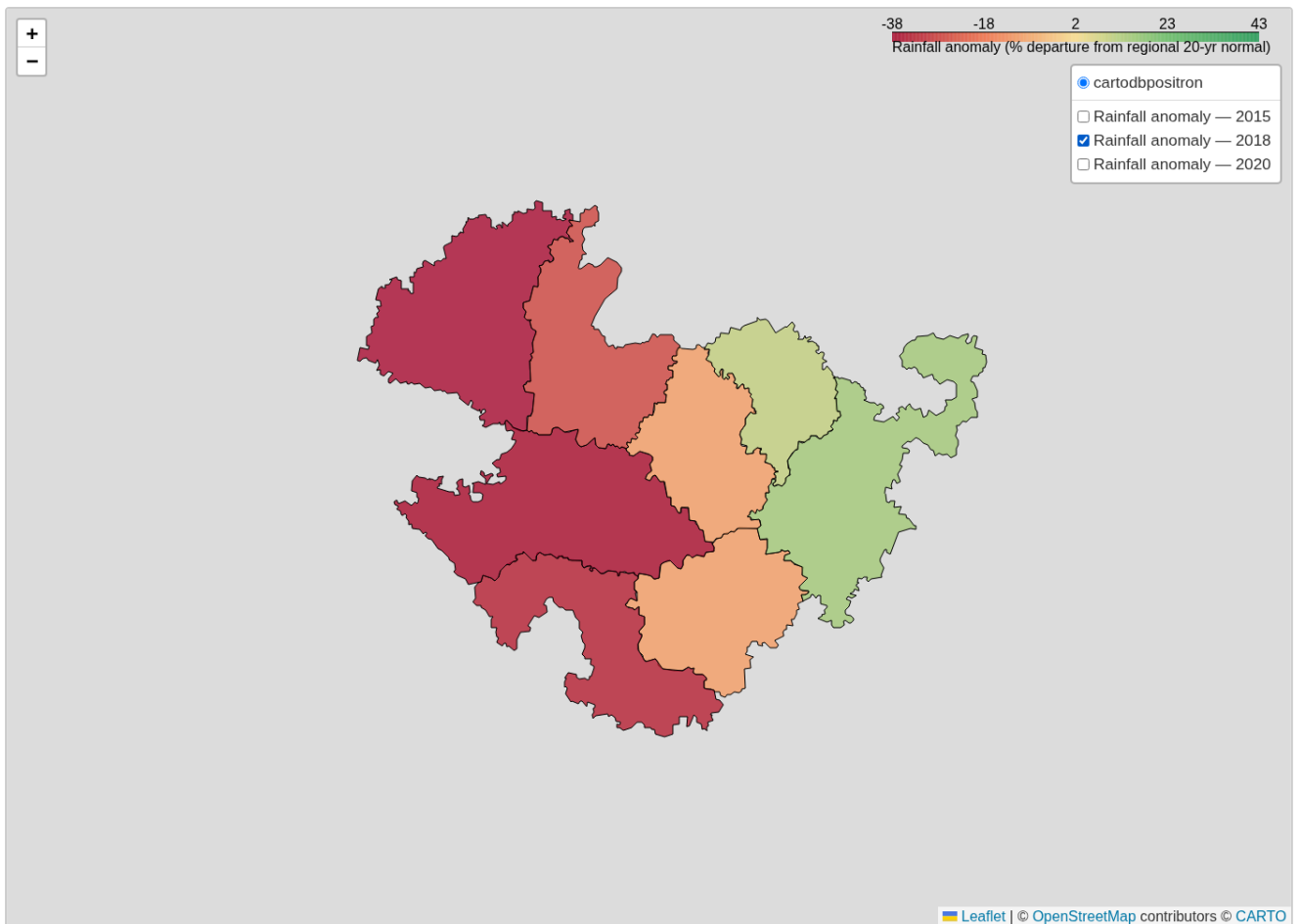


Figure 13b — Rainfall anomaly by district, 2018 (drought)

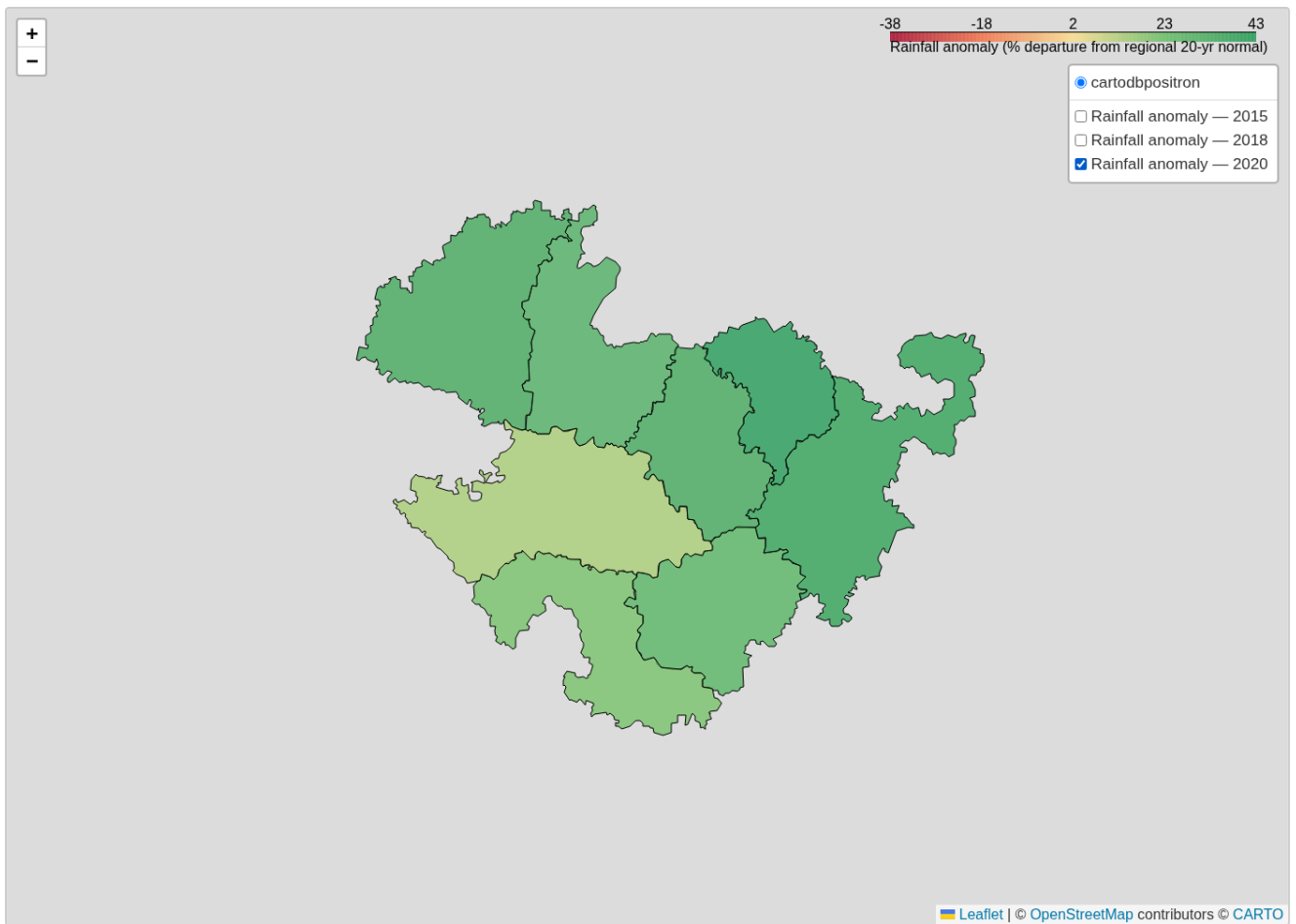


Figure 13c — Rainfall anomaly by district, 2020 (normal)

8. Combined SIF and Rainfall Comparison

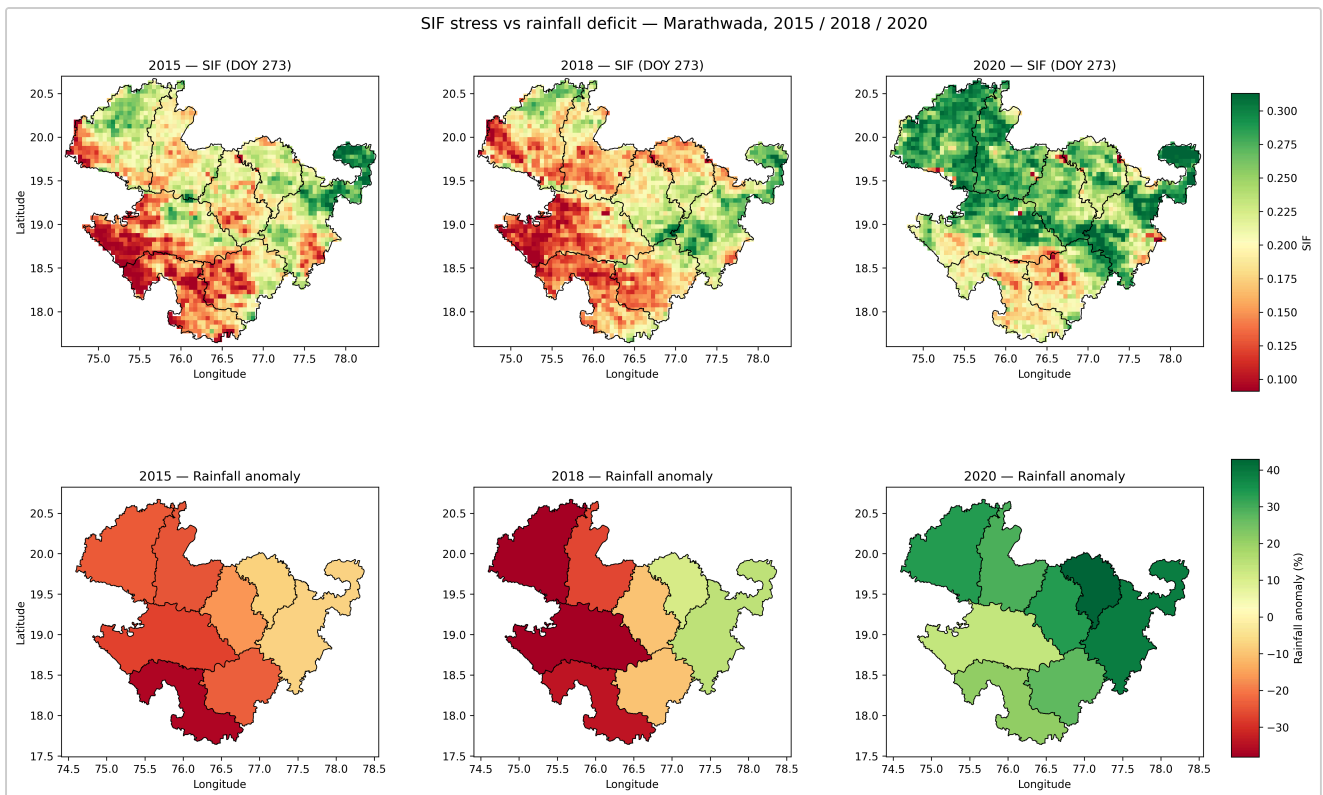


Figure 14

SIF (top row) and rainfall anomaly (bottom row) by district, 2015/2018/2020, presented jointly for direct spatial comparison.

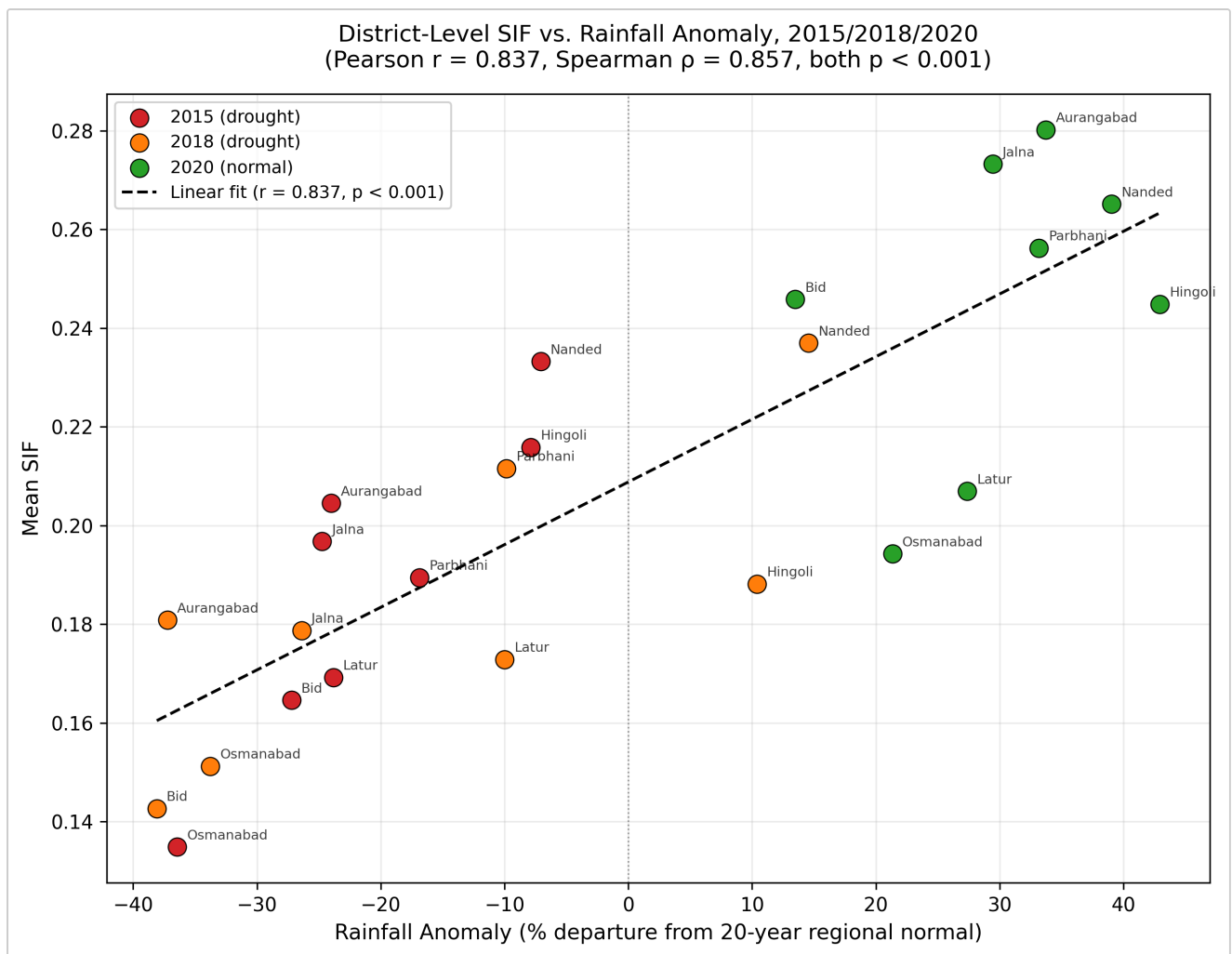


Figure 15

District-level mean SIF plotted against rainfall anomaly (%), all 24 district-year observations, with a linear fit (Pearson $r = 0.837$, $p < 0.001$).